

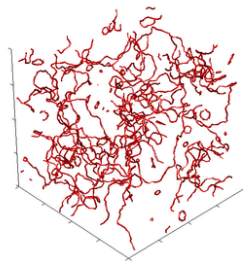
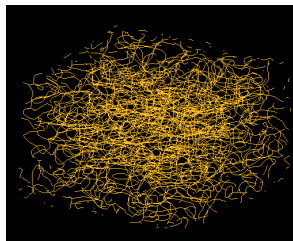
Hydrodynamic Approaches to String Networks

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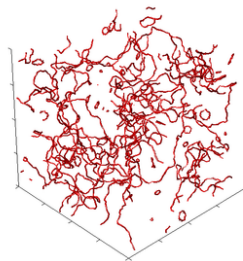
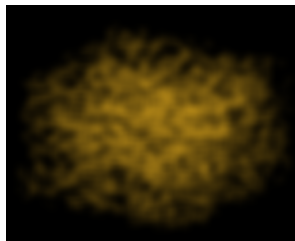
String Networks

- Examples
 - ▶ Cosmic Strings
 - ▶ Quantum Turbulence
 - ▶ Magnetic Flux Tubes
- Different scales
- In superfluid, HVBK equations
- Apply idea to other systems
- Outline
 - ▶ Relativistic perfect fluids
 - ▶ Coarse-grained Nambu-Goto fluid
 - ▶ Variational principles
 - ▶ Dissipative effects



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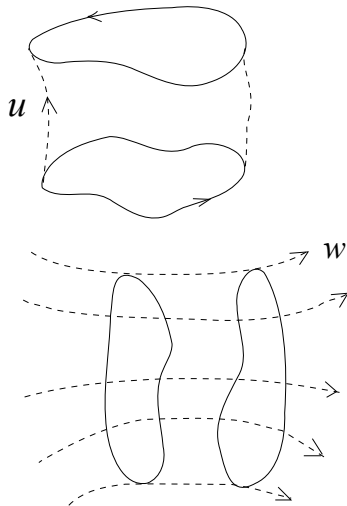
Perfect Fluid

- $T^{\mu\nu} = (\rho + p)u^\mu u^\nu - pg^{\mu\nu}$
- Equation of state: $\rho(n_a)$ is a function of extensive currents
- $\nabla_\mu(n_a u^\mu) \equiv \nabla_\mu n_a^\mu = 0$
- Chemical potentials: $\mu^a \equiv \frac{\partial \rho}{\partial n_a}$
- $\rho = -p + \mu^a n_a$
- Legendre transform: $n_a = \frac{\partial p}{\partial \mu^a}$

$$\begin{aligned}
 0 &= \nabla_\mu T^\mu_\nu = \nabla_\mu (\mu^a n_a u^\mu u_\nu - p \delta^\mu_\nu) \\
 &= n_a^\mu \nabla_\mu (\mu^a u_\nu) - n_a \nabla_\nu \mu^a \\
 &= 2 n_a^\mu \nabla_{[\mu} \mu^a_{\nu]}
 \end{aligned}$$

Vorticity

- $\omega_{\mu\nu} \equiv 2 \nabla_{[\mu} \mu_{\nu]}$
- $n^\mu \omega_{\mu\nu} = 0$
- Exact differential $d\omega = 0$
- Kelvin circulation theorem
- Spacelike direction $w^\mu \omega_{\mu\nu} = 0$ (Simple bivector)
- Conserved flux
- Field lines of w as strings



F Tensor

- The dual $F^{\mu\nu} \equiv \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}\omega_{\rho\sigma}$ is tangent to field lines
- $\nabla_\mu F^{\mu\nu} = 0$
- $F^{\mu\nu} \equiv \begin{pmatrix} 0 & -B_1 & -B_2 & -B_3 \\ B_1 & 0 & E_3 & -E_2 \\ B_2 & -E_3 & 0 & E_1 \\ B_3 & E_2 & -E_1 & 0 \end{pmatrix}$
- $\nabla \cdot \mathbf{B} = 0, \quad \dot{\mathbf{B}} = -\nabla \times \mathbf{E}$
- In MHD, flux tubes carry magnetic flux rather than angular momentum
- Just as $n^\mu \equiv nu^\mu$ describes charge, $F^{\mu\nu} \equiv \varphi\Sigma^{\mu\nu}$ describes flux
- String fluid: φ itself is thermodynamic quantity

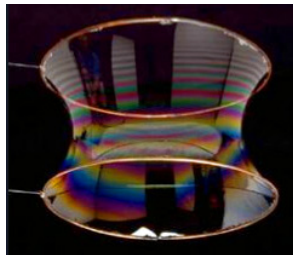
Individual Strings

- Consider ‘microscopic’ string network
- Worldsheet in spacetime $X^\mu(\tau, \sigma)$
- Velocity and tangent vector

$$U^\mu \equiv X^\mu_{,\tau}$$

$$W^\mu \equiv X^\mu_{,\sigma}$$

- Coordinate and gauge invariance
- Nambu-Goto strings
 - ▶ Worldlines extremize ‘length’
 - ▶ Worldsheets extremize spacetime area
 - ▶ Generalize to current carrying strings



Coarse-Grained Currents

- Nambu-Goto Action $-\int d^2\eta \sqrt{-h}$, where $h_{ab} \equiv g_{\mu\nu} X_{,a}^\mu X_{,b}^\nu$
- Energy-momentum tensor over spacetime

$$T^{\mu\nu}(x)\sqrt{-g} = \int d^2\eta \sqrt{-h} h^{ab} X_{,a}^\mu X_{,b}^\nu \delta(x - X(\eta))$$

- Energy-momentum coarse-grained in volume $\Delta V(x)$

$$\langle T \rangle^{\mu\nu}(x) = \frac{1}{\Delta V} \int T^{\mu\nu}(x') \sqrt{-g} d^4x'$$

- Singular conserved current has form

$$\int d^2\eta J^a X_{,a}^\mu \delta(x - X(\eta))$$

J is current on worldsheet, $\partial_a J^a = 0$

- For instance, $\partial_a(\sqrt{-h} h^{ab} X_{,b}^\nu) = 0$

F Tensor Again

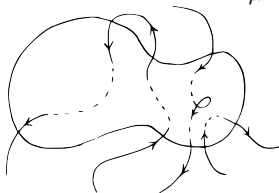
- Regardless of dynamics, currents on worldsheet

$$J^{a\nu} \equiv \epsilon^{ab} X_{,b}^{\nu}, \quad \partial_a J^{a\nu} = 0$$

- Corresponds to an antisymmetric tensor in spacetime

$$F^{\mu\nu} \sqrt{-g} = \int d^2\eta J^{a\nu} X_{,a}^{\mu} \delta(x - X(\eta))$$

- Coarse-grained $\langle F \rangle$ plays same role as vorticity or EM field tensors
- The conservation $\nabla_{\mu} F^{\mu\nu} = 0$ related to topological flux of strings



Light-Cone Coordinate Vectors

- A perturbation travels along the string in two directions
- The tangent vectors to these paths in spacetime are denoted A, B

$$A^\mu = U^\mu - W^\mu, \quad B^\mu = U^\mu + W^\mu$$

- $\langle T \rangle^{\mu\nu} = \frac{1}{\Delta V} \int d\rho A^{(\mu} B^{\nu)}, \quad \langle F \rangle^{\mu\nu} = \frac{1}{\Delta V} \int d\rho A^{[\mu} B^{\nu]}$
- Coarse-graining $\sim$ expectation value with respect to energy-density measure
- Simplifying assumption that statistics of A and B are independent. 'Local equilibrium'
- Kinetic theory of string segments

Equations of Motion

- $\langle T \rangle^{\mu\nu} = \rho \bar{A}^{(\mu} \bar{B}^{\nu)}, \quad \langle F \rangle^{\mu\nu} = \rho \bar{A}^{[\mu} \bar{B}^{\nu]}$
- From $\nabla_\mu \langle T \rangle^{\mu\nu} = \nabla_\mu \langle F \rangle^{\mu\nu} = 0$, in terms of $\bar{U}, \bar{W}$

$$\nabla_\mu (\rho \bar{U}^\mu) = \nabla_\mu (\rho \bar{W}^\mu) = 0$$

$$\bar{U}^\mu \nabla_\mu \bar{U}^\nu - \bar{W}^\mu \nabla_\mu \bar{W}^\nu = 0$$

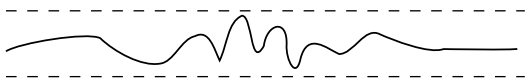
$$\bar{W}^\mu \nabla_\mu \bar{U}^\nu - \bar{U}^\mu \nabla_\mu \bar{W}^\nu = 0$$

- If $\bar{W} = 0$, dust of small loops.
- Frobenius theorem implies integrable submanifolds.
- Implied by $\langle F \rangle$ being a simple bivector. Not in dissipative case.
- Submanifolds as worldsheets?



Current-Carrying Strings

- Submanifolds are wiggly string worldsheets



- Mass-per-length and tension $MT = \mu_0^2$
- In Carter's approach treat as perfect fluid confined to worldsheet, with some conserved current ν
- Equation of state $M(\nu) = \mu_0 \sqrt{1 + \nu^2}$
- Tension defined through $M = T + \mu\nu$, $\mu \equiv \frac{\partial M}{\partial \nu}$
- More general strings by changing form of $M(\nu)$
- There is necessarily a second 'dual' conserved current
- What is the 'wobble density' ν ?
- If $\nu = (T_H/\mu_0)s_\nu$, temperature $T \rightarrow T_H$

Fluid Lagrangians

- Taub variational principle,

$$\mathcal{L} = -\rho(n), \quad n^2 = -\frac{1}{3!} \tilde{n}^{\lambda\mu\nu} \tilde{n}_{\lambda\mu\nu}$$

- Only works for constrained variations $\delta\tilde{n}$
- But for string fluid we find $\rho = \varphi M(\nu)$, where $\varphi^2 = \frac{1}{2} \tilde{F}^{\mu\nu} \tilde{F}_{\mu\nu}$
- Considering units, take $n = \varphi\nu$
- So $\mathcal{L} = -\rho = -\sqrt{(\mu_0\varphi)^2 + (T_H n)^2}$, which leads to correct $T^{\mu\nu}$
- Constructing $\mathcal{L}$ from different $M(\nu)$ leads to different submanifolds
- If n dependence vanishes, $\mathcal{L} = -\mu_0\sqrt{\varphi^2}$, is Stachel-Letelier model
- Different dependence on φ leads to a pressure term in the energy momentum tensor

Perfect String Fluid

- There is a conserved charge and flux $\nabla_\mu n^\mu = \nabla_\mu F^{\mu\nu} = 0$
- Velocity u^μ is unit vector $\frac{1}{n}n^\mu$
- Tangent vector w^μ is unit vector $\frac{1}{\varphi}F^{\mu\nu}u_\nu$
- Given an arbitrary Lagrangian $\mathcal{L}(\varphi^2, n^2)$ we then find

$$T^{\mu\nu} = (\rho + p)u^\mu u^\nu - (\tau + p)w^\mu w^\nu - pg^{\mu\nu}$$

- Pressure defined so that

$$\rho = -p + mn + \mu\varphi,$$

with $m = -\mathcal{L}_{,n}$ and $\mu = -\mathcal{L}_{,\varphi}$

- String tension $\tau \equiv \rho - mn$

Magnetohydrodynamics

- F is dual of EM field tensor
- Simple bivector condition $u^\mu \tilde{F}_{\mu\nu} = 0$ equivalent to 'frozen-in' magnetic field condition
- Full fluid sum of ordinary fluid and EM parts

$$\begin{aligned}
 T^{\mu\nu} &= T_{\text{m}}^{\mu\nu} + T_{\text{EM}}^{\mu\nu} \\
 &= (\rho + p)u^\mu u^\nu - pg^{\mu\nu} - \tilde{F}^{\mu\rho} \tilde{F}^\nu_\rho + \frac{1}{4}g^{\mu\nu} \tilde{F}^{\rho\sigma} \tilde{F}_{\rho\sigma} \\
 &= (\rho + p + \varphi^2)u^\mu u^\nu - \varphi^2 w^\mu w^\nu - (p + \frac{1}{2}\varphi^2)g^{\mu\nu}
 \end{aligned}$$

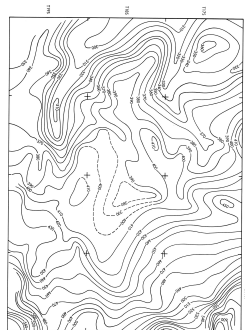
- Choosing the Lagrangian $\mathcal{L} = -\rho - \frac{1}{2}\varphi = -\rho_{\text{total}}$,

$$p_{\text{total}} = p + \frac{1}{2}\varphi^2,$$

and this is indeed a 'perfect string fluid'

2+1d Field

- What are the fields that are varied?
- 'Height' function X such that contour lines are field lines w
- $F = \star dX \Rightarrow \nabla_\mu F^{\mu\nu} = 0$
- If $\mathcal{L} = -\sqrt{-dX^2}$, contour lines move as Nambu-Goto strings
- Different dependencies on dX^2 lead to pressure
- e.g. Massless scalar field $\mathcal{L} = g^{\mu\nu} \partial_\mu X \partial_\nu X$
- In 3+1d, describes domain walls



Variational Principle

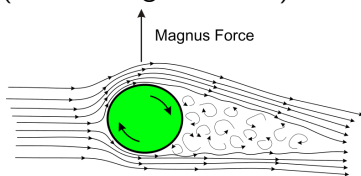
- In 3+1d need two fields X, Y to label submanifolds
- $dX \wedge dY$ annihilates tangent space of submanifold
- X, Y normalized so that $\tilde{F} \equiv dX \wedge dY$ is flux two-form
- $\nabla_\mu F^{\mu\nu} = 0$ by construction
- To describe fluid particle worldlines need an additional label Z
- Velocity confined to integrable submanifolds by construction
- Normalized so that $\tilde{n} \equiv dX \wedge dY \wedge dZ$ number density volume form
- $\mathcal{L}(\frac{1}{2}\tilde{F}^2, -\frac{1}{3!}\tilde{n}^2)$ varied by $X, Y, Z \dots$

$$-\frac{3}{2}F^{\lambda\mu}\nabla_{[\kappa}(\mu\Sigma_{\lambda\mu]) + 2n^\lambda\nabla_{[\kappa}(mu_{\lambda]}) = 0.$$

- Can reintroduce dependence on $-dX^2$ to describe strings, particles confined to domain walls

Relabeling Symmetries

- X, Y, Z only appear in combinations $\tilde{F}, \tilde{n}$
- $Z \rightarrow Z + F(X, Y)$ symmetry. Carter's dual current
- Symmetries of X, Y , symplectic transformations
- Effective field theory picture (Dubovsky et al), higher derivative terms
- Instead of dependence on $\tilde{n}$, $mu^\mu = dZ' + XdY$ leads to superfluid description
- $dX \wedge dY \wedge dZ'$ also allowed by symmetry, describes Zhukovsky lift (Carter-Langlois model)



- Dissipation by integrating out irrelevant degrees of freedom

Dissipative Fluids Introduction

- Equation of state already involves entropy $\rho = -p + Ts + \mu\varphi$
- Equations of motion imply conservation of entropy

$$u_\nu \nabla_\mu T^{\mu\nu} = T \nabla_\mu (s u^\mu) + \mu (\nabla_\mu \varphi u^\mu - \varphi h_\mu^\lambda \nabla_\lambda u^\mu) = 0$$

- But when T, F are more general, extra terms. Constrained by 2nd Law
- Still true $\nabla_\mu T^{\mu\nu} = \nabla_\mu F^{\mu\nu} = \nabla_\mu n^\mu = 0$, but no longer perfect fluid form
- No longer a uniquely preferred velocity. Eckart vs Landau-Lifshitz frames
- Alternatively, choose u, w from eigenspace of $F^{\lambda\rho} F_{\rho\mu} \sim h_\mu^\lambda$. String frame vs integrable frame
- If nearly perfect fluid, all frames are close. Principle of frame invariance

Dissipative Terms

- Decompose into parts parallel and orthogonal to u, w

$$T^{\rho\sigma} = T_{\text{eq}}^{\rho\sigma} + 2u^{(\rho}q^{\sigma)} + \dots$$

$$F^{\rho\sigma} = F_{\text{eq}}^{\rho\sigma} + 2w^{[\rho}\nu^{\sigma]} + \dots$$

- Now, from $T^{-1}u_\nu\nabla_\mu T^{\mu\nu} = 0$

$$\begin{aligned} \nabla_\mu (su^\mu + \frac{1}{T}q^\mu - \frac{\mu}{T}\nu^\mu) - q^\nu (\nabla_\nu \frac{1}{T} + \frac{1}{T}u^\mu \nabla_\mu u_\nu) \\ + \nu^\nu (\nabla_\nu \frac{\mu}{T} + \frac{\mu}{T}w^\mu \nabla_\mu w_\nu) + \dots = 0 \end{aligned}$$

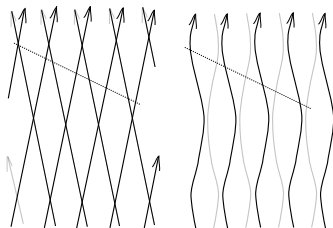
- From 1st Law, $ds = \frac{1}{T}d\rho - \frac{\mu}{T}d\varphi$
- From 2nd Law,

$$q_T^\nu = \kappa_T \perp^{\mu\nu} (\nabla_\mu T - T u^\nu \nabla_\nu u_\mu),$$

$$\nu^\mu = \xi_T \perp^{\mu\rho} (\nabla_\rho \frac{\mu}{T} + \frac{\mu}{T} w^\sigma \nabla_\sigma w_\rho)$$

Entropy Production

- Non-relativistic limit, $\nu = -\xi_T \left(\nabla_{\perp} \frac{\mu}{T} - \frac{\mu}{T} (\mathbf{w} \cdot \nabla) \mathbf{w} \right)$
- Field lines want to go to lower μ/T
- Entropy due to string curvature $\mathbf{w} \times \nu = \xi_T \frac{\mu}{T} (\nabla \times \mathbf{w})_{\perp}$
- Another term with distinct coefficient $\tilde{\mathbf{G}} = \xi_L \frac{\mu}{T} (\nabla \times \mathbf{w})_{\parallel}$



Resistive Magnetohydrodynamics

- In MHD, $\varphi \mathbf{w} = \mathbf{B}$,
- From $\nabla_\mu F^{\mu i} = 0$, Faraday's Law

$$\dot{\mathbf{B}} = -\nabla \times (\mathbf{B} \times \mathbf{v}) - \nabla \times (\mathbf{w} \times \nu) - \nabla \times \tilde{\mathbf{G}}$$

- $\mathbf{E} = -\mathbf{v} \times \mathbf{B} + \frac{\xi}{T}(\nabla \times \mathbf{B}) + \frac{\xi_T}{T^2} \mathbf{B} \times \nabla T$
- By Ampere's Law, $\nabla \times \mathbf{B} \sim \mathbf{J}$. So just Ohm's Law with $\sigma = T\xi^{-1}$
- Temperature gradient term, Nernst effect

Higher-Order Effects

- Longitudinal heat flow $Q_L = \kappa_L w^\mu (\nabla_\mu T - T u^\nu \nabla_\nu u_\mu)$
- Appearance of heat equation $\dot{T} = \frac{\kappa_L}{C} \partial_w^2 T$
- But entropy current s is a function of T , F expanded about an equilibrium state
- Including second-order terms, e.g. $-\frac{1}{2} k u^\mu Q_L^2$

$$k C T^2 \ddot{T} + \frac{C}{\kappa_L} \dot{T} = \partial_w^2 T$$

- s should not depend on the frame it is expanded about. This restricts form of second-order terms
- Q_L transforms differently from other quantities, so second-order term is fixed
- Speed of second sound, $c_s = \sqrt{\frac{\tau}{\rho}} = \sqrt{1 - \left(\frac{T}{T_H}\right)^2}$